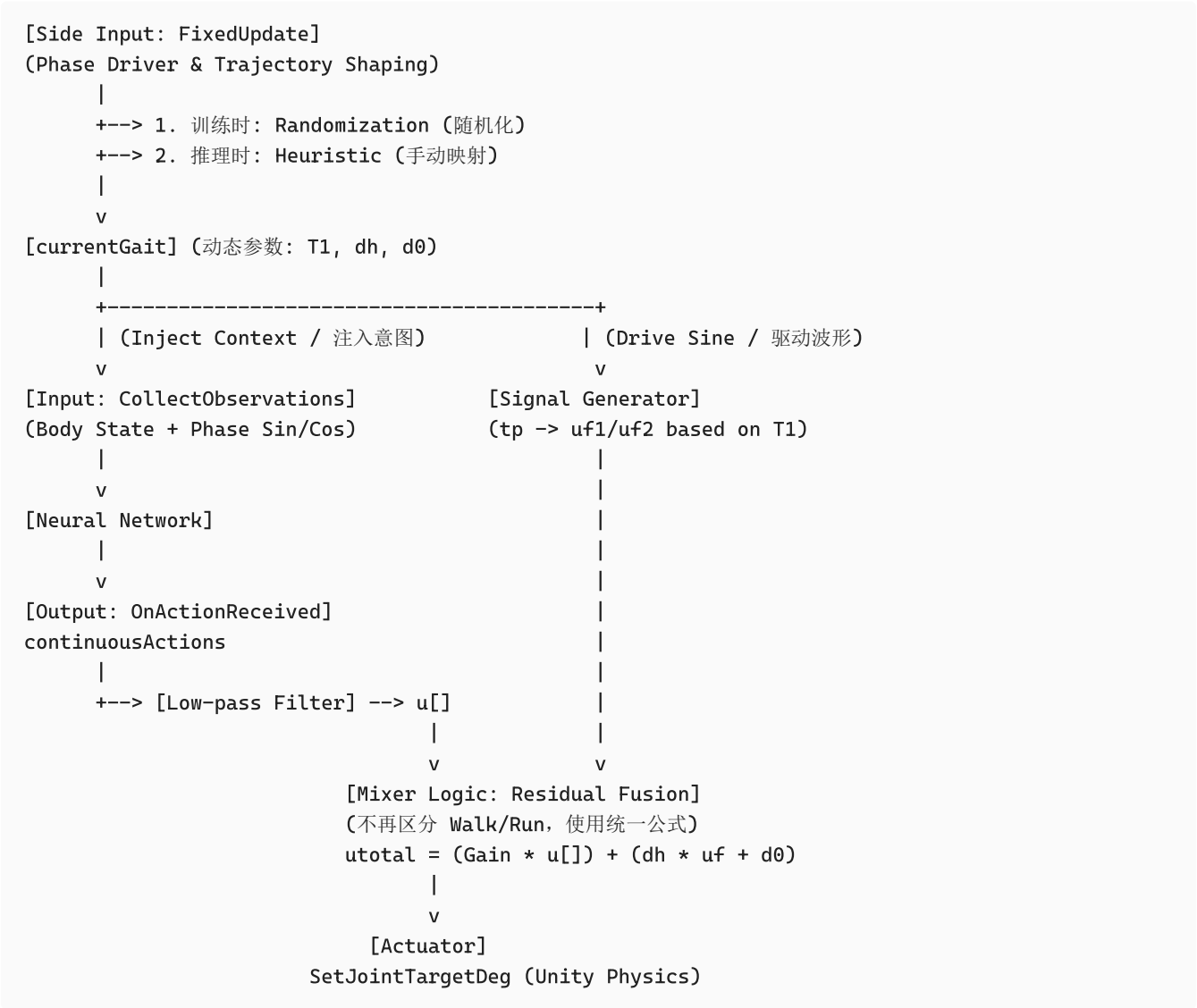


技术报告_Continuous Gait Transition (RL)

1. 技术报告：MoB 架构下的代码重构与实现

为了实现双足机器人从“Baseline：离散步态切换”升级到“连续步态切换”的能力，我将原有的有限状态机（FSM）逻辑重构为基于 **Multiplicity of Behavior (MoB)** 的参数化控制架构。

数据流



1.1 数据结构重构：从“状态枚举”到“参数向量”

原状 (Baseline): 可以发现Baseline中控制run和walk两种步态的核心为步态核心超参数组合（T1, dh, d0）；使用 enum StyleB { walk, run } 进行硬编码切换。变量 T1, dh, d0 散落在类中，通过 if-else 赋值。

修改 (MoB): 引入结构体封装，将步态视为一个连续的参数向量。

• 代码实现：

```
// [New] 定义步态参数结构体
[System.Serializable]
public struct GaitParams
{
    public float T1; // 周期/频率
    public float dh; // 振幅/抬腿高度
    public float d0; // 偏置/姿态高度
}
```

```
// [New] 引入当前状态与目标状态，用于插值
[SerializeField] private GaitParams currentGait; // 当前执行目标（平滑/过渡）
public GaitParams targetGait; // 遥控指令目标
[SerializeField] private int changeGaitTimer = 0; // 训练计时器
```

- 在 FixedUpdate() 中：

```
// 参数平滑插值，避免物理突变
currentGait.T1 = Mathf.Lerp(currentGait.T1, targetGait.T1, 0.1f);
currentGait.dh = Mathf.Lerp(currentGait.dh, targetGait.dh, 0.1f);
currentGait.d0 = Mathf.Lerp(currentGait.d0, targetGait.d0, 0.1f);
```

1.2 感知空间增强：注入“相位”与“意图”

原状 (Baseline): 神经网络仅观测物理状态（速度、角度）。它是“盲目”的，不知道当前脚本希望它快跑还是慢走，全靠脚本的物理强制力拖拽。

修改 (MoB): 在 CollectObservations 中显式注入**相位信息**和**任务上下文**，使 RL 大脑与脊髓节律同步。

- 代码实现：

```
public override void CollectObservations(VectorSensor sensor)
{
    // ... 原有物理观测 ...

    // [New 1] 相位编码 (Phase Encoding)
    // 使用 Sin/Cos 解决 0 与 2π 的不连续问题，告诉 NN 当前腿摆在什么位置
    float phase = (float)tp / (2f * Mathf.Max(1f, currentGait.T1));
    sensor.AddObservation(Mathf.Sin(2 * Mathf.PI * phase));
    sensor.AddObservation(Mathf.Cos(2 * Mathf.PI * phase));

    // [New 2] 任务上下文 (Context/Command)
    // 告诉 NN 当前的目标风格（是跑还是走？）
    sensor.AddObservation(currentGait.T1);
    sensor.AddObservation(currentGait.dh);
    sensor.AddObservation(currentGait.d0);
}
```

注意：需同步在 Unity Inspector 中增加 Observation Space Size (+5)。

1.3 控制逻辑统一：移除“硬编码状态机”

原状 (Baseline):

```
if (Walk) { T1=30; utotal[...] += ... }
if (Run) { T1=20; utotal[...] += ... } // 逻辑割裂，无法混合
```

修改 (MoB): 统一使用同一套残差公式，通过 currentGait 变量驱动。无论走还是跑，物理本质都是正弦波驱动，区别仅在于参数值不同。

- 代码实现 (OnActionReceived):

```
// [Refactor] 移除所有 if (Walk/Run) 分支，使用统一逻辑
// 动态获取参数
T1 = (int)currentGait.T1;
dh = currentGait.dh;
d0 = currentGait.d0;

// 执行统一的混合公式 (Residual Fusion)
```

```
// idx[0] 为髋关节
utotal[idx[0]] += (dh * uf1 + d0) * Mathf.Sign(idx[0]);
// ... 其他关节同理 ...

// [Crucial] 移除“自杀逻辑”
// Delete: if (target != last) EndEpisode();
// 允许在同一回合内动态改变参数，训练过渡能力
```

1.4 训练策略升级：域随机化 (Domain Randomization)

原状 (Baseline): 静态训练。设为 Walk 就一直练 Walk。

修改 (MoB): 在训练过程中，动态、随机地改变步态参数(T1,dh,d0)，迫使 Agent 适应参数的动态变化。

train: 课程学习训练：从慢走 (40,10,0) 到快跑 (25,40,20) 的所有参数组合可能组合

```
if (train)
{
    changeGaitTimer++;
    if (changeGaitTimer > 300) // 每3秒变一次
    {
        changeGaitTimer = 0;
        RandomizeGaitCommand_Smoothed();
    }
    // 平滑插值
    currentGait.T1 = Mathf.Lerp(currentGait.T1, targetGait.T1, 0.1f);
    currentGait.dh = Mathf.Lerp(currentGait.dh, targetGait.dh, 0.1f);
    currentGait.d0 = Mathf.Lerp(currentGait.d0, targetGait.d0, 0.1f);
}
```

```
void RandomizeGaitCommand_Smoothed()
{
    int step = Academy.Instance.StepCount;
    // 计算课程进度 (0.0 到 1.0)，在 200万步时完成
    float progress = Mathf.Clamp01((float)step / 2000000f);
    // 动态范围：随着训练深入，难度逐渐加大
    float minT1 = Mathf.Lerp(38f, 24f, progress);
    float maxDh = Mathf.Lerp(14f, 41f, progress); // 从小碎步(14) 扩展到 大跨步(41)
    float maxD0 = Mathf.Lerp(5f, 20f, progress); // 从直立(5) 扩展到 蹲姿(20)
    targetGait.T1 = Random.Range(minT1, 41f);
    targetGait.dh = Random.Range(9f, maxDh);
    targetGait.d0 = Random.Range(0f, maxD0);
}
```

测试：

```
changeGaitTimer++;
// 起点：慢走
const float startT1 = 40f;
const float startDh = 10f;
const float startD0 = 0f;

// 终点：快跑
const float endT1 = 25f;
const float endDh = 40f;
const float endD0 = 20f;

// 每 ... 前进 10%
float progress = Mathf.Clamp01(changeGaitTimer / 3000f);
targetGait.T1 = Mathf.Lerp(startT1, endT1, progress);
```

```
targetGait.dh = Mathf.Lerp(startDh, endDh, progress);
targetGait.d0 = Mathf.Lerp(startD0, endD0, progress);

currentGait.T1 = Mathf.Lerp(currentGait.T1, targetGait.T1, 0.1f);
currentGait.dh = Mathf.Lerp(currentGait.dh, targetGait.dh, 0.1f);
currentGait.d0 = Mathf.Lerp(currentGait.d0, targetGait.d0, 0.1f);
```

2. 实现过程

其他设置：

1. Unity界面，观察数量+5（改成38）
2. Unity界面，maxstep修改成5000并在cs文件把 FixedUpdate() 中 `tt>=1000` 注释掉。（原代码中step=1000后机器人会重置）

```
if (Mathf.Abs(EulerTrans(body.eulerAngles[0])) > 20f ||
    Mathf.Abs(EulerTrans(body.eulerAngles[2])) > 20f ) // || tt>=1000
```

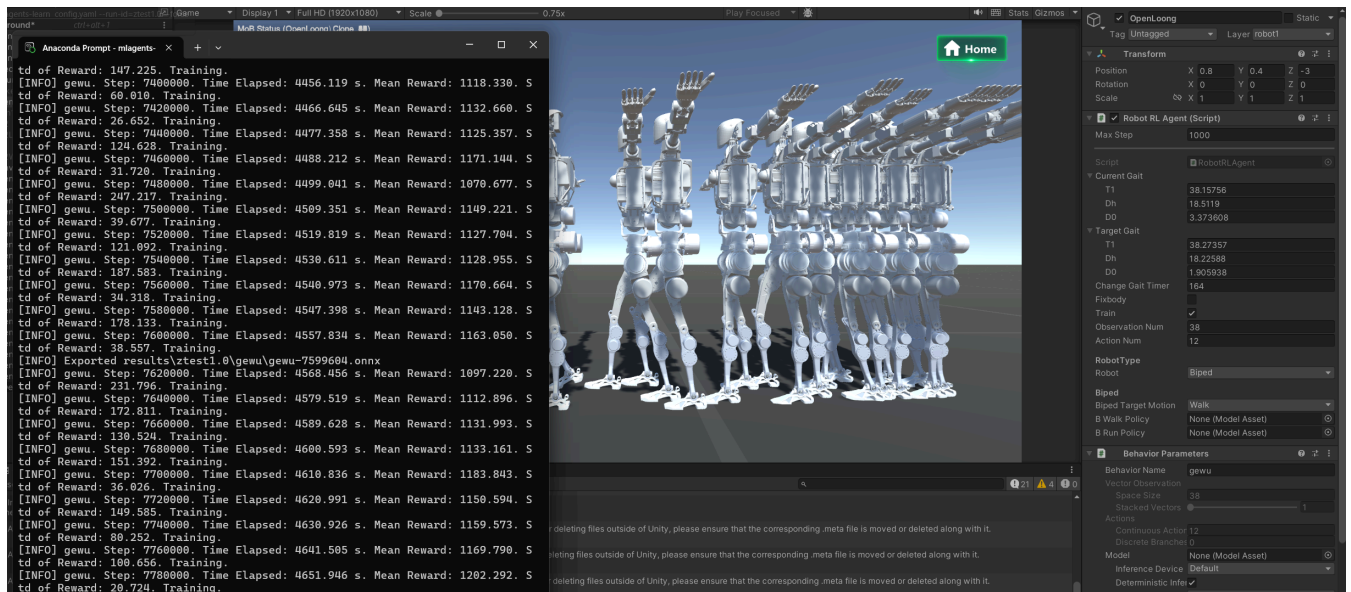
3. Unity界面，训练场地增大10倍（不然一定距离后机器人会掉下去）
4. 训练时，cs文件内，机器人克隆数量修改成20（根据cpu性能决定）

训练过程

fixbody 测试底层正弦波（见附件视频）

训练700wstep时，已经可以平稳行走（可能不用700w就可以了，中间睡了一觉没看到...）

此时（T1, Dh, D0）的取值为（38.15,18.51,3.37）



此时，删除源代码中的步数限制，修改maxstep为5000

一开始reward还在提升

```
INFO] gewu. Step: 900000. Time Elapsed: 53.768 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 902000. Time Elapsed: 67.342 s. Mean Reward: 6799.005. Std of Reward: 352.663. Training.
INFO] gewu. Step: 904000. Time Elapsed: 74.776 s. Mean Reward: 6853.167. Std of Reward: 35.943. Training.
INFO] gewu. Step: 906000. Time Elapsed: 92.243 s. Mean Reward: 6969.277. Std of Reward: 0.000. Training.
INFO] gewu. Step: 908000. Time Elapsed: 102.709 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 910000. Time Elapsed: 113.101 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 912000. Time Elapsed: 119.611 s. Mean Reward: 7280.069. Std of Reward: 75.030. Training.
INFO] gewu. Step: 914000. Time Elapsed: 133.614 s. Mean Reward: 7259.951. Std of Reward: 35.708. Training.
INFO] gewu. Step: 916000. Time Elapsed: 144.125 s. Mean Reward: 7564.449. Std of Reward: 0.000. Training.
INFO] gewu. Step: 918000. Time Elapsed: 154.557 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 920000. Time Elapsed: 165.058 s. Mean Reward: 5531.707. Std of Reward: 0.000. Training.
INFO] Exported results\ztest1.0\gewu\gewu-9199080.onnx
INFO] gewu. Step: 922000. Time Elapsed: 175.584 s. Mean Reward: 7667.285. Std of Reward: 91.259. Training.
INFO] gewu. Step: 924000. Time Elapsed: 182.511 s. Mean Reward: 7687.806. Std of Reward: 35.576. Training.
INFO] gewu. Step: 926000. Time Elapsed: 199.288 s. Mean Reward: 7756.780. Std of Reward: 0.000. Training.
INFO] gewu. Step: 928000. Time Elapsed: 212.415 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 930000. Time Elapsed: 223.340 s. Mean Reward: 7061.743. Std of Reward: 798.611. Training.
INFO] gewu. Step: 932000. Time Elapsed: 233.874 s. Mean Reward: 8004.603. Std of Reward: 44.871. Training.
INFO] gewu. Step: 934000. Time Elapsed: 245.249 s. Mean Reward: 4511.321. Std of Reward: 3606.991. Training.
INFO] gewu. Step: 936000. Time Elapsed: 255.875 s. Mean Reward: 4324.568. Std of Reward: 3340.112. Training.
INFO] gewu. Step: 938000. Time Elapsed: 262.523 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 940000. Time Elapsed: 275.363 s. Mean Reward: 8135.579. Std of Reward: 0.000. Training.
INFO] gewu. Step: 942000. Time Elapsed: 285.979 s. Mean Reward: 8254.636. Std of Reward: 50.240. Training.
INFO] gewu. Step: 944000. Time Elapsed: 296.510 s. Mean Reward: 8251.211. Std of Reward: 7.516. Training.
INFO] gewu. Step: 946000. Time Elapsed: 307.063 s. Mean Reward: 8331.047. Std of Reward: 22.749. Training.
INFO] gewu. Step: 948000. Time Elapsed: 317.543 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 950000. Time Elapsed: 327.947 s. Mean Reward: 8492.319. Std of Reward: 0.000. Training.
INFO] gewu. Step: 952000. Time Elapsed: 338.183 s. Mean Reward: 8551.509. Std of Reward: 47.751. Training.
INFO] gewu. Step: 954000. Time Elapsed: 348.794 s. Mean Reward: 8588.429. Std of Reward: 37.391. Training.
INFO] gewu. Step: 956000. Time Elapsed: 359.167 s. Mean Reward: 7247.954. Std of Reward: 2470.574. Training.
INFO] gewu. Step: 958000. Time Elapsed: 365.659 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 960000. Time Elapsed: 377.984 s. Mean Reward: 8740.586. Std of Reward: 0.000. Training.
INFO] Exported results\ztest1.0\gewu\gewu-9599284.onnx
INFO] gewu. Step: 962000. Time Elapsed: 388.337 s. Mean Reward: 8852.071. Std of Reward: 33.114. Training.
INFO] gewu. Step: 964000. Time Elapsed: 399.743 s. Mean Reward: 3929.574. Std of Reward: 4088.117. Training.
INFO] gewu. Step: 966000. Time Elapsed: 410.343 s. Mean Reward: 8909.518. Std of Reward: 38.178. Training.
INFO] gewu. Step: 968000. Time Elapsed: 420.733 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 970000. Time Elapsed: 431.058 s. Mean Reward: 9052.534. Std of Reward: 0.000. Training.
INFO] gewu. Step: 972000. Time Elapsed: 441.476 s. Mean Reward: 9025.901. Std of Reward: 21.754. Training.
INFO] gewu. Step: 974000. Time Elapsed: 451.873 s. Mean Reward: 9010.826. Std of Reward: 11.766. Training.
INFO] gewu. Step: 976000. Time Elapsed: 462.915 s. Mean Reward: 7026.510. Std of Reward: 2753.876. Training.
INFO] gewu. Step: 978000. Time Elapsed: 473.321 s. No episode was completed since last summary. Training.
INFO] gewu. Step: 980000. Time Elapsed: 485.124 s. Mean Reward: 9005.274. Std of Reward: 0.000. Training.
INFO] gewu. Step: 982000. Time Elapsed: 495.489 s. Mean Reward: 9013.756. Std of Reward: 100.764. Training.
INFO] gewu. Step: 984000. Time Elapsed: 501.922 s. Mean Reward: 9029.551. Std of Reward: 39.373. Training.
INFO] gewu. Step: 986000. Time Elapsed: 512.315 s. Mean Reward: 9016.672. Std of Reward: 73.591. Training.
INFO] gewu. Step: 988000. Time Elapsed: 522.814 s. No episode was completed since last summary. Training.
```

后来reward一直在震荡

```
[INFO] gewu. Step: 10260000. Time Elapsed: 726.433 s. Mean Reward: 5989.549. Std of Reward: 4225.803. Training.
[INFO] gewu. Step: 10280000. Time Elapsed: 732.579 s. No episode was completed since last summary. Training.
[INFO] gewu. Step: 10300000. Time Elapsed: 742.731 s. Mean Reward: 8859.429. Std of Reward: 76.345. Training.
[INFO] gewu. Step: 10320000. Time Elapsed: 753.364 s. Mean Reward: 8646.715. Std of Reward: 482.198. Training.
[INFO] gewu. Step: 10340000. Time Elapsed: 763.762 s. Mean Reward: 6339.404. Std of Reward: 3725.447. Training.
[INFO] gewu. Step: 10360000. Time Elapsed: 774.273 s. Mean Reward: 8787.036. Std of Reward: 6.038. Training.
[INFO] gewu. Step: 10380000. Time Elapsed: 784.445 s. Mean Reward: 8694.826. Std of Reward: 0.000. Training.
[INFO] gewu. Step: 10400000. Time Elapsed: 794.842 s. Mean Reward: 8419.134. Std of Reward: 595.608. Training.
[INFO] Exported results\ztest1.0\gewu\gewu-10399029.onnx
[INFO] gewu. Step: 10420000. Time Elapsed: 805.616 s. Mean Reward: 8642.537. Std of Reward: 117.025. Training.
[INFO] gewu. Step: 10440000. Time Elapsed: 815.807 s. Mean Reward: 8118.068. Std of Reward: 723.883. Training.
[INFO] gewu. Step: 10460000. Time Elapsed: 826.083 s. No episode was completed since last summary. Training.
[INFO] gewu. Step: 10480000. Time Elapsed: 836.193 s. Mean Reward: 8384.656. Std of Reward: 436.974. Training.
[INFO] gewu. Step: 10500000. Time Elapsed: 845.737 s. Mean Reward: 8575.158. Std of Reward: 131.641. Training.
[INFO] gewu. Step: 10520000. Time Elapsed: 856.103 s. Mean Reward: 8578.365. Std of Reward: 133.616. Training.
[INFO] gewu. Step: 10540000. Time Elapsed: 866.750 s. Mean Reward: 7053.823. Std of Reward: 2577.212. Training.
[INFO] gewu. Step: 10560000. Time Elapsed: 876.978 s. No episode was completed since last summary. Training.
[INFO] gewu. Step: 10580000. Time Elapsed: 887.403 s. Mean Reward: 7423.811. Std of Reward: 1548.951. Training.
[INFO] gewu. Step: 10600000. Time Elapsed: 898.305 s. Mean Reward: 8190.464. Std of Reward: 768.723. Training.
[INFO] gewu. Step: 10620000. Time Elapsed: 909.149 s. Mean Reward: 8510.784. Std of Reward: 49.510. Training.
[INFO] gewu. Step: 10640000. Time Elapsed: 919.584 s. Mean Reward: 8370.254. Std of Reward: 0.000. Training.
[INFO] gewu. Step: 10660000. Time Elapsed: 925.363 s. Mean Reward: 8476.953. Std of Reward: 1.707. Training.
[INFO] gewu. Step: 10680000. Time Elapsed: 935.721 s. Mean Reward: 7825.118. Std of Reward: 1871.182. Training.
[INFO] gewu. Step: 10700000. Time Elapsed: 946.877 s. Mean Reward: 8529.773. Std of Reward: 55.589. Training.
[INFO] gewu. Step: 10720000. Time Elapsed: 957.601 s. Mean Reward: 8495.993. Std of Reward: 101.629. Training.
[INFO] gewu. Step: 10740000. Time Elapsed: 968.021 s. Mean Reward: 4725.000. Std of Reward: 0.000. Training.
[INFO] gewu. Step: 10760000. Time Elapsed: 978.096 s. Mean Reward: 8347.758. Std of Reward: 152.332. Training.
[INFO] gewu. Step: 10780000. Time Elapsed: 988.897 s. Mean Reward: 8393.204. Std of Reward: 77.880. Training.
[INFO] gewu. Step: 10800000. Time Elapsed: 998.874 s. Mean Reward: 7202.482. Std of Reward: 1867.077. Training.
[INFO] Exported results\ztest1.0\gewu\gewu-10799972.onnx
[INFO] gewu. Step: 10820000. Time Elapsed: 1008.802 s. Mean Reward: 7025.290. Std of Reward: 1295.464. Training.
[INFO] gewu. Step: 10840000. Time Elapsed: 1019.158 s. Mean Reward: 6492.049. Std of Reward: 1881.482. Training.
[INFO] gewu. Step: 10860000. Time Elapsed: 1029.570 s. Mean Reward: 8257.318. Std of Reward: 93.070. Training.
[INFO] gewu. Step: 10880000. Time Elapsed: 1040.775 s. Mean Reward: 7272.925. Std of Reward: 2490.185. Training.
[INFO] gewu. Step: 10900000. Time Elapsed: 1051.112 s. Mean Reward: 8364.204. Std of Reward: 95.182. Training.
[INFO] gewu. Step: 10920000. Time Elapsed: 1061.461 s. Mean Reward: 8382.539. Std of Reward: 48.061. Training.
[INFO] gewu. Step: 10940000. Time Elapsed: 1071.811 s. Mean Reward: 8334.292. Std of Reward: 100.701. Training.
[INFO] gewu. Step: 10960000. Time Elapsed: 1082.414 s. Mean Reward: 6884.312. Std of Reward: 2567.693. Training.
[INFO] gewu. Step: 10980000. Time Elapsed: 1093.615 s. Mean Reward: 8281.877. Std of Reward: 22.148. Training.
[INFO] gewu. Step: 11000000. Time Elapsed: 1104.046 s. Mean Reward: 8284.510. Std of Reward: 29.649. Training.
[INFO] gewu. Step: 11020000. Time Elapsed: 1114.375 s. Mean Reward: 8219.717. Std of Reward: 26.920. Training.
[INFO] gewu. Step: 11040000. Time Elapsed: 1123.480 s. Mean Reward: 8290.493. Std of Reward: 33.288. Training.
[INFO] gewu. Step: 11060000. Time Elapsed: 1134.120 s. Mean Reward: 6091.167. Std of Reward: 2788.203. Training.
[INFO] gewu. Step: 11080000. Time Elapsed: 1140.510 s. Mean Reward: 8237.737. Std of Reward: 0.000. Training.
[INFO] gewu. Step: 11100000. Time Elapsed: 1150.735 s. Mean Reward: 5404.010. Std of Reward: 3024.406. Training.
[INFO] gewu. Step: 11120000. Time Elapsed: 1161.852 s. Mean Reward: 8251.549. Std of Reward: 150.886. Training.
[INFO] gewu. Step: 11140000. Time Elapsed: 1172.220 s. Mean Reward: 8232.226. Std of Reward: 73.128. Training.
```

2000w时甚至不如1000w的效果好

```
[INFO] gewu. Step: 19660000. Time Elapsed: 6332.471 s. Mean Reward: 6516.613. Std of Reward: 1358.130. Training.
[INFO] gewu. Step: 19680000. Time Elapsed: 6345.273 s. Mean Reward: 2363.978. Std of Reward: 8626.309. Training.
[INFO] gewu. Step: 19700000. Time Elapsed: 6356.802 s. Mean Reward: 4047.751. Std of Reward: 3425.321. Training.
[INFO] gewu. Step: 19720000. Time Elapsed: 6369.393 s. Mean Reward: 6410.186. Std of Reward: 1965.984. Training.
[INFO] gewu. Step: 19740000. Time Elapsed: 6382.580 s. Mean Reward: 5756.792. Std of Reward: 2901.428. Training.
[INFO] gewu. Step: 19760000. Time Elapsed: 6394.787 s. Mean Reward: 5963.594. Std of Reward: 1900.832. Training.
[INFO] gewu. Step: 19780000. Time Elapsed: 6406.860 s. Mean Reward: 5814.098. Std of Reward: 2242.048. Training.
[INFO] gewu. Step: 19800000. Time Elapsed: 6419.641 s. Mean Reward: 7241.996. Std of Reward: 382.071. Training.
[INFO] gewu. Step: 19820000. Time Elapsed: 6432.084 s. Mean Reward: 6860.187. Std of Reward: 1017.515. Training.
[INFO] gewu. Step: 19840000. Time Elapsed: 6444.789 s. Mean Reward: 7453.436. Std of Reward: 48.440. Training.
[INFO] gewu. Step: 19860000. Time Elapsed: 6451.835 s. Mean Reward: 7269.826. Std of Reward: 343.084. Training.
[INFO] gewu. Step: 19880000. Time Elapsed: 6464.941 s. Mean Reward: 6433.353. Std of Reward: 2038.925. Training.
[INFO] gewu. Step: 19900000. Time Elapsed: 6478.352 s. Mean Reward: 5535.952. Std of Reward: 2544.286. Training.
[INFO] gewu. Step: 19920000. Time Elapsed: 6489.706 s. Mean Reward: 6537.033. Std of Reward: 2476.751. Training.
[INFO] gewu. Step: 19940000. Time Elapsed: 6502.408 s. Mean Reward: 6895.301. Std of Reward: 1316.098. Training.
[INFO] gewu. Step: 19960000. Time Elapsed: 6514.967 s. Mean Reward: 7514.332. Std of Reward: 219.757. Training.
[INFO] gewu. Step: 19980000. Time Elapsed: 6527.230 s. Mean Reward: 5960.319. Std of Reward: 2848.877. Training.
[INFO] gewu. Step: 20000000. Time Elapsed: 6539.863 s. Mean Reward: 5621.945. Std of Reward: 2913.801. Training.
[INFO] Exported results\ztest1.0\gewu\gewu-19999403.onnx
[INFO] Exported results\ztest1.0\gewu\gewu-20000403.onnx
```

最终训练2000w的神经网络测试（见附件视频）。在(T1,dh,d0)分别于（40，32），（10，25），（0，10）区间内变化时，表现还好。但步频和步幅上去之后就摔了。